Mathematics
for
Information Technology

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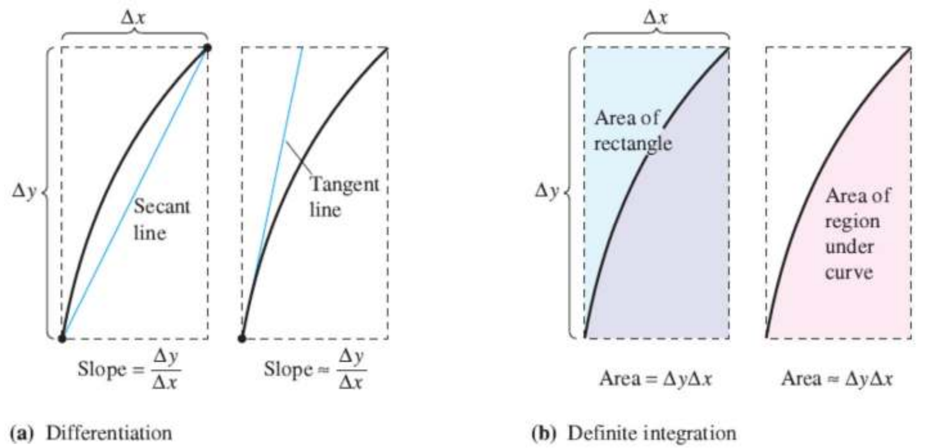
INTEGRATION (II)

The Fundamental Theorem of Calculus III

Objectives

- Evaluate a definite integral using the Fundamental Theorem of Calculus.
- Understand and use the Mean Value Theorem for Integrals.
- Find the average value of a function over a closed interval.
- Understand and use the Second Fundamental Theorem of Calculus.
- Understand and use the Net Change Theorem.

The Fundamental Theorem of Calculus



Differentiation and definite integration have an "inverse" relationship.

THEOREM 4.9 THE FUNDAMENTAL THEOREM OF CALCULUS

If a function f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on the interval $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

GUIDELINES FOR USING THE FUNDAMENTAL THEOREM OF CALCULUS

1. *Provided you can find* an antiderivative of f, you now have a way to evaluate a definite integral without having to use the limit of a sum.
2. When applying the Fundamental Theorem of Calculus, the following notation is convenient.

$$\boxed{\begin{aligned}\int_a^b f(x)\,dx &= F(x)\Big]_a^b \\ &= F(b) - F(a)\end{aligned}}$$

For instance, to evaluate $\int_1^3 x^3\,dx$, you can write

$$\int_1^3 x^3\,dx = \frac{x^4}{4}\Big]_1^3 = \frac{3^4}{4} - \frac{1^4}{4} = \frac{81}{4} - \frac{1}{4} = 20.$$

3. It is not necessary to include a constant of integration C in the antiderivative because

$$\begin{aligned}\int_a^b f(x)\,dx &= \Big[F(x) + C\Big]_a^b \\ &= [F(b) + C] - [F(a) + C] \\ &= F(b) - F(a).\end{aligned}$$

EXAMPLE 1 Evaluating a Definite Integral

Evaluate each definite integral.

a. $\int_1^2 (x^2 - 3)\,dx$ **b.** $\int_1^4 3\sqrt{x}\,dx$ **c.** $\int_0^{\pi/4} \sec^2 x\,dx$

Solution

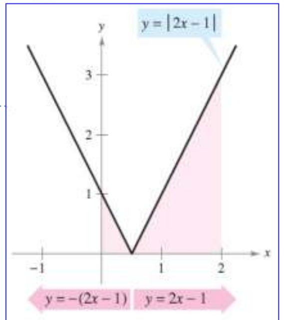
a. $\int_1^2 (x^2 - 3)\,dx =$

b. $\int_1^4 3\sqrt{x}\,dx =$

c. $\int_0^{\pi/4} \sec^2 x\,dx =$

EXAMPLE 2 A Definite Integral Involving Absolute Value

Evaluate $\int_0^2 |2x - 1|\,dx.$



The definite integral of y on $[0, 2]$ is $\frac{5}{2}$.

Solution

$$|2x-1| = \begin{cases} -(2x-1), & x < \frac{1}{2} \\ 2x-1, & x \geq \frac{1}{2} \end{cases}$$

$$\int_0^2 |2x-1|\,dx = \int_0^{1/2} -(2x-1)\,dx + \int_{1/2}^2 (2x-1)\,dx$$

$$= \left[-x^2+x\right]_0^{1/2} + \left[x^2-x\right]_{1/2}^2$$

$$= \left(-\frac{1}{4}+\frac{1}{2}\right) - (0+0) + (4-2) - \left(\frac{1}{4}-\frac{1}{2}\right)$$

$$= \frac{5}{2}$$

EXAMPLE 3 Using the Fundamental Theorem to Find Area

Find the area of the region bounded by the graph of $y = 2x^2 - 3x + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 2$, as shown in Figure

?

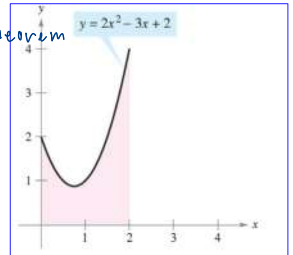
Note! $y > 0$ on the interval $[0, 2]$.

$$\text{Area} = \int_0^2 (2x^2 - 3x + 2)\,dx \quad \longrightarrow \text{Integrate between } x=0 \text{ and } x=2$$

$$= \left[\frac{2x^3}{3} - \frac{3x^2}{2} + 2x\right]_0^2 \quad \text{find antiderivative}$$

$$= \left(\frac{16}{3} - 6 + 4\right) - (0 - 0 + 0) \quad \text{Apply fundamental Theorem}$$

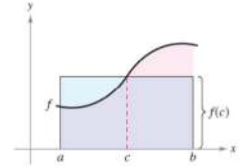
$$= \frac{10}{3} \quad \text{simplify}$$



The area of the region bounded by the graph of y, the x-axis, $x = 0$, and $x = 2$ is $\frac{10}{3}$.

The Mean Value Theorem for Integrals

Mean Value Theorem for Integrals states that somewhere "between" the inscribed and circumscribed rectangles there is a rectangle whose area is precisely equal to the area of the region under the curve, as shown in Figure



Mean value rectangle:
$f(c)(b-a) = \int_a^b f(x)\,dx$

THEOREM 4.10 MEAN VALUE THEOREM FOR INTEGRALS

If f is continuous on the closed interval $[a, b]$, then there exists a number c in the closed interval $[a, b]$ such that

$$\int_a^b f(x)\,dx = f(c)(b-a).$$

NOTE Notice that Theorem 4.10 does not specify how to determine c. It merely guarantees the existence of at least one number c in the interval.

Average Value of a Function

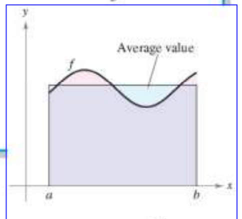
The value of $f(c)$ given in the Mean Value Theorem for Integrals is called the **average value** of f on the interval $[a, b]$.

DEFINITION OF THE AVERAGE VALUE OF A FUNCTION ON AN INTERVAL

If f is integrable on the closed interval $[a, b]$, then the **average value** of f on the interval is

$$\frac{1}{b-a}\int_a^b f(x)\,dx. \quad \text{\# ค่าเฉลี่ยของฟังก์ชัน}$$

$$\frac{\text{หาพท.ใต้กราฟ}}{\text{ความกว้าง}}$$



Average value $= \frac{1}{b-a}\int_a^b f(x)\,dx$

NOTE Notice in Figure that the area of the region under the graph of f is equal to the area of the rectangle whose height is the average value.

EXAMPLE 4 Finding the Average Value of a Function

function ในช่วง $[1, 4]$ มีค่าเฉลี่ย = ?

Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[1, 4]$.

Solution The average value is given by

?

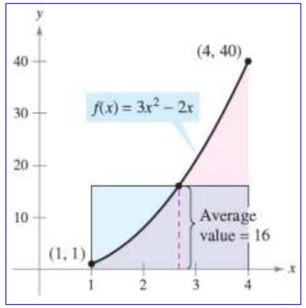
$$\frac{1}{b-a}\int_a^b f(x)\,dx = \frac{1}{4-1}\int_1^4 (3x^2 - 2x)\,dx$$

$$= \frac{1}{3}\left[x^3 - x^2\right]_1^4 \quad \text{เอา 4 ไปแทนใน } x \quad 1 : \text{constant เอามาลบ}$$

$$= \frac{1}{3}\left[64 - 16 - (1 - 1)\right]$$

$$= \frac{48}{3}$$

$$= 16$$



The Second Fundamental Theorem of Calculus

The Definite Integral as a Number

Constant

$$\int_a^b f(x)\,dx$$

Constant

f is a function of x.

The Definite Integral as a Function of x

F is a function of x.

$$F(x) = \int_a^x f(t)\,dt$$

Constant

f is a function of t.

EXAMPLE 6 The Definite Integral as a Function

Evaluate the function

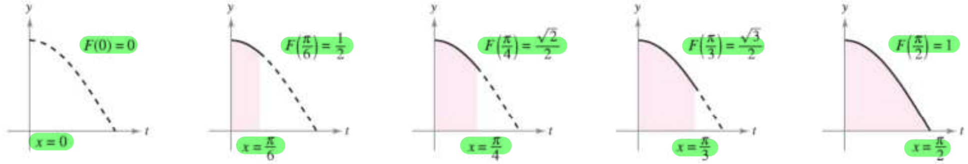
$$F(x) = \int_0^x \cos t\,dt$$

at $x = 0$, $\pi/6$, $\pi/4$, $\pi/3$, and $\pi/2$.

Solution You could evaluate five different definite integrals, one for each of the given upper limits. However, it is much simpler to fix x (as a constant) temporarily to obtain

$$\int_0^x \cos t\,dt = \sin t\Big]_0^x = \sin x - \sin 0 = \sin x.$$

Now, using $F(x) = \sin x$, you can obtain the results shown in Figure



$F(x) = \int_0^x \cos t\,dt$ is the area under the curve $f(t) = \cos t$ from 0 to x.

an **accumulation function**

accumulating the area

THEOREM 4.11 THE SECOND FUNDAMENTAL THEOREM OF CALCULUS

If f is continuous on an open interval I containing a, then, for every x in the interval,

$$\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x).$$

constant

same variable

It can be used to avoid the long process of integration followed by differentiation

EXAMPLE 8 Using the Second Fundamental Theorem of Calculus

Find the derivative of $F(x) = \int_{\pi/2}^{x^3} \cos t\,dt$.

Solution Using $u = x^3$, you can apply the Second Fundamental Theorem of Calculus with the Chain Rule as shown.

$F'(x) = \dfrac{dF}{du}\dfrac{du}{dx}$	Chain Rule
$= \dfrac{d}{du}[F(x)]\dfrac{du}{dx}$	Definition of $\dfrac{dF}{du}$
$= \dfrac{d}{du}\left[\int_{\pi/2}^{x^3} \cos t\,dt\right]\dfrac{du}{dx}$	Substitute $\int_{\pi/2}^{x^3} \cos t\,dt$ for $F(x)$.
$= \dfrac{d}{du}\left[\int_{\pi/2}^{u} \cos t\,dt\right]\dfrac{du}{dx}$	Substitute u for x^3.
$= (\cos u)(3x^2)$	Apply Second Fundamental Theorem of Calculus.
$= (\cos x^3)(3x^2)$	Rewrite as function of x. ■

$$\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$$

Alternative

Because the integrand in Example 8 is easily integrated, you can verify the derivative as follows.

$$F(x) = \int_{\pi/2}^{x^3} \cos t\,dt = \sin t\Big]_{\pi/2}^{x^3} = \sin x^3 - \sin\frac{\pi}{2} = (\sin x^3) - 1$$

In this form, you can apply the Power Rule to verify that the derivative is the same as that obtained in Example 8.

$$F'(x) = (\cos x^3)(3x^2)$$

Net Change Theorem

THEOREM 4.12 THE NET CHANGE THEOREM

The definite integral of the rate of change of a quantity $F'(x)$ gives the total change, or **net change,** in that quantity on the interval $[a, b]$.

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$ Net change of F #หาผลรวมของก.เปลี่ยนแปลง

EXAMPLE 9 Using the Net Change Theorem

A chemical flows into a storage tank at a rate of $180 + 3t$ liters per minute, where $0 \le t \le 60$. Find the amount of the chemical that flows into the tank during the first 20 minutes.

Solution Let $c(t)$ be the amount of the chemical in the tank at time t. Then $c'(t)$ represents the rate at which the chemical flows into the tank at time t. During the first 20 minutes, the amount that flows into the tank is

20

วินาทีที่ 0

$$\int_0^{20} c'(t)\,dt = \int_0^{20} (180 + 3t)\,dt$$

$$= \left[180t + \frac{3}{2}t^2\right]_0^{20}$$ 20 แทนใน t

$$= 3600 + 600 = 4200.$$

So, the amount that flows into the tank during the first 20 minutes is 4200 liters.

EXAMPLE 10 Solving a Particle Motion Problem

A particle is moving along a line so that its velocity is $v(t) = t^3 - 10t^2 + 29t - 20$ feet per second at time t.

a. What is the displacement of the particle on the time interval $1 \le t \le 5$?

b. What is the total distance traveled by the particle on the time interval $1 \le t \le 5$?

Solution

a. displacement

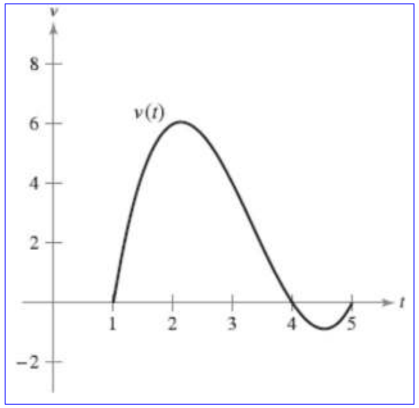
$$\int_1^5 v(t)\,dt = \int_1^5 (t^3 - 10t^2 + 29t - 20)\,dt$$

$$= \left[\frac{t^4}{4} - \frac{10}{3}t^3 + \frac{29}{2}t^2 - 20t\right]_1^5$$

$$= \frac{25}{12} - \left(-\frac{103}{12}\right)$$

$$= \frac{128}{12}$$

$$= \frac{32}{3}$$



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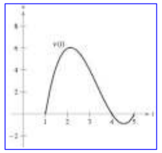
a. What is the displacement of the particle on the time interval $1 \le t \le 5$?

b. What is the total distance traveled by the particle on the time interval $1 \le t \le 5$?

Solution



b. $\int_1^5 |v(t)|\,dt$. v can be factored as $(t-1)(t-4)(t-5)$

, $v(t) \ge 0$ on $[1, 4]$ and $v(t) \le 0$ on $[4, 5]$

$$\int_1^5 |v(t)|\,dt = \int_1^4 v(t)\,dt - \int_4^5 v(t)\,dt$$

$$= \int_1^4 (t^3 - 10t^2 + 29t - 20)\,dt - \int_4^5 (t^3 - 10t^2 + 29t - 20)\,dt$$

$$= \left[\frac{t^4}{4} - \frac{10}{3}t^3 + \frac{29}{2}t^2 - 20t\right]_1^4 - \left[\frac{t^4}{4} - \frac{10}{3}t^3 + \frac{29}{2}t^2 - 20t\right]_4^5$$

$$= \frac{45}{4} - \left(-\frac{7}{12}\right)$$

$$= \frac{71}{6} \text{ feet.}$$

BREAK

Integration by Substitution

IV

Objectives

- Use pattern recognition to find an indefinite integral.
- Use a change of variables to find an indefinite integral.
- Use the General Power Rule for Integration to find an indefinite integral.
- Use a change of variables to evaluate a definite integral.

Pattern Recognition

In this section you will study techniques for integrating composite functions. The discussion is split into two parts—*pattern recognition* and *change of variables*. Both techniques involve a ***u*-substitution.** With pattern recognition you perform the substitution mentally, and with change of variables you write the substitution steps.

THEOREM 4.13 ANTIDIFFERENTIATION OF A COMPOSITE FUNCTION

Let g be a function whose range is an interval I, and let f be a function that is continuous on I. If g is differentiable on its domain and F is an antiderivative of f on I, then

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C.$$

Letting $u = g(x)$ gives $du = g'(x)\,dx$ and

$$\int f(u)\,du = F(u) + C.$$

NOTE The statement of Theorem 4.13 doesn't tell how to distinguish between $f(g(x))$ and $g'(x)$ in the integrand. As you become more experienced at integration, your skill in doing this will increase. Of course, part of the key is familiarity with derivatives.

Outside function

$$\int f(g(x))\underbrace{g'(x)}\,dx = F(g(x)) + C$$

Inside function

Derivative of inside function

EXAMPLE 1 Recognizing the $f(g(x))g'(x)$ Pattern

Find $\int (x^2 + 1)^2(2x)\, dx$.

Solution Letting $g(x) = x^2 + 1$, you obtain

$$g'(x) = 2x$$

and

$$f(g(x)) = f(x^2 + 1) = (x^2 + 1)^2.$$

From this, you can recognize that the integrand follows the $f(g(x))g'(x)$ pattern. Using the Power Rule for Integration and Theorem 4.13, you can write

$$\int \overbrace{(x^2 + 1)^2}^{f(g(x))}\overbrace{(2x)}^{g'(x)}\, dx = \frac{1}{3}(x^2 + 1)^3 + C.$$

Try using the Chain Rule to check that the derivative of $\frac{1}{3}(x^2 + 1)^3 + C$ is the integrand of the original integral.

Constant Multiple Rule

$$\int kf(x)\, dx = k\int f(x)\, dx.$$

Many integrands contain the essential part (the variable part) of $g'(x)$ but are missing a constant multiple. In such cases, you can multiply and divide by the necessary constant multiple, as shown in Example 3.

EXAMPLE 3 Multiplying and Dividing by a Constant

EXAMPLE 1

Find $\int (x^2 + 1)^2(2x)\, dx$.

Find $\int x(x^2 + 1)^2\, dx$.

Solution This is similar to the integral given in Example 1, except that the integrand is missing a factor of 2. Recognizing that $2x$ is the derivative of $x^2 + 1$, you can let $g(x) = x^2 + 1$ and supply the $2x$ as follows.

$$\begin{aligned}
\int x(x^2 + 1)^2\, dx &= \int (x^2 + 1)^2\left(\frac{1}{2}\right)(2x)\, dx && \text{Multiply and divide by 2.}\\
&= \frac{1}{2}\int \overbrace{(x^2 + 1)^2}^{f(g(x))}\,\overbrace{(2x)}^{g'(x)}\, dx && \text{Constant Multiple Rule}\\
&= \frac{1}{2}\left[\frac{(x^2 + 1)^3}{3}\right] + C && \text{Integrate.}\\
&= \frac{1}{6}(x^2 + 1)^3 + C && \text{Simplify.}
\end{aligned}$$

EXAMPLE 3 Multiplying and Dividing by a Constant

Find $\int x(x^2 + 1)^2\, dx$.

Solution

In practice, most people would not write as many steps as are shown in Example 3. For instance, you could evaluate the integral by simply writing

$$\begin{aligned}
\int x(x^2 + 1)^2\, dx &= \frac{1}{2}\int (x^2 + 1)^2\, 2x\, dx\\
&= \frac{1}{2}\left[\frac{(x^2 + 1)^3}{3}\right] + C\\
&= \frac{1}{6}(x^2 + 1)^3 + C.
\end{aligned}$$

Change of Variables

With a formal **change of variables,** you completely rewrite the integral in terms of u and du (or any other convenient variable). Although this procedure can involve more written steps than the pattern recognition illustrated in Examples 1 to 3, it is useful for complicated integrands. The change of variables technique uses the Leibniz notation for the differential. That is, if $u = g(x)$, then $du = g'(x)\,dx$, and the integral in Theorem 4.13 takes the form

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du = F(u) + C.$$

EXAMPLE 4 Change of Variables

Find $\int \sqrt{2x - 1}\,dx$.

Solution First, let u be the inner function, $u = 2x - 1$. Then calculate the differential du to be $du = 2\,dx$. Now, using $\sqrt{2x-1} = \sqrt{u}$ and $dx = du/2$, substitute to obtain

$$\begin{aligned}
\int \sqrt{2x-1}\,dx &= \int \sqrt{u}\left(\frac{du}{2}\right) && \text{Integral in terms of } u\\
&= \frac{1}{2}\int u^{1/2}\,du && \text{Constant Multiple Rule}\\
&= \frac{1}{2}\left(\frac{u^{3/2}}{3/2}\right) + C && \text{Antiderivative in terms of } u\\
&= \frac{1}{3}u^{3/2} + C && \text{Simplify.}\\
&= \frac{1}{3}(2x-1)^{3/2} + C. && \text{Antiderivative in terms of } x
\end{aligned}$$

EXAMPLE 5 Change of Variables

Find $\int x\sqrt{2x - 1}\,dx$.

$u = 2x - 1 \longrightarrow x = (u+1)/2$ Solve for x term of u

$$\begin{aligned}
\int x\sqrt{2x-1}\,dx &= \int \left(\frac{u+1}{2}\right) u^{1/2}\left(\frac{du}{2}\right)\\
&= \frac{1}{4}\int (u^{3/2} + u^{1/2})\,du\\
&= \frac{1}{4}\left(\frac{u^{5/2}}{5/2} + \frac{u^{3/2}}{3/2}\right) + C\\
&= \frac{1}{10}(2x-1)^{5/2} + \frac{1}{6}(2x-1)^{3/2} + C.
\end{aligned}$$

GUIDELINES FOR MAKING A CHANGE OF VARIABLES

1. Choose a substitution $u = g(x)$. Usually, it is best to choose the *inner* part of a composite function, such as a quantity raised to a power.
2. Compute $du = g'(x)\,dx$.
3. Rewrite the integral in terms of the variable u.
4. Find the resulting integral in terms of u.
5. Replace u by $g(x)$ to obtain an antiderivative in terms of x.
6. Check your answer by differentiating.

The General Power Rule for Integration

THEOREM 4.14 THE GENERAL POWER RULE FOR INTEGRATION

If g is a differentiable function of x, then

$$\int [g(x)]^n g'(x)\,dx = \frac{[g(x)]^{n+1}}{n+1} + C, \quad n \neq -1.$$

Equivalently, if $u = g(x)$, then

$$\int u^n\,du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1.$$

EXAMPLE 7 Substitution and the General Power Rule

a. $$\int 3(3x-1)^4\,dx = \int \overbrace{(3x-1)^4}^{u^4}\overbrace{(3)\,dx}^{du} = \overbrace{\frac{(3x-1)^5}{5}}^{u^5/5} + C$$

b. $$\int (2x+1)(x^2+x)\,dx = \int \overbrace{(x^2+x)^1}^{u^1}\overbrace{(2x+1)}^{du}\,dx = \overbrace{\frac{(x^2+x)^2}{2}}^{u^2/2} + c$$

c. $$\int 3x^2\sqrt{x^3-2}\,dx = \int \overbrace{(x^3-2)^{1/2}}^{u^{1/2}}\overbrace{(3x^2)}^{du}\,dx = \overbrace{\frac{(x^3-2)^{3/2}}{3/2}}^{u^{3/2}/(3/2)} + c = \frac{2}{3}(x^3-2)^{3/2} + c$$

d. $$\int \frac{-4x}{(1-2x^2)^2}\,dx = \int \overbrace{(1-2x^2)^{-2}}^{u^{-2}}\overbrace{(-4x)}^{du}\,dx = \overbrace{\frac{(1-2x^2)^{-1}}{-1}}^{u^{-1}/(-1)} + c = -\frac{1}{1-2x^2} + c$$

e. $$\int \cos^2 x \sin x\,dx = \int \overbrace{(\cos x)^2}^{u^2}\overbrace{(-\sin x)}^{du}\,dx = -\overbrace{\frac{(\cos x)^3}{3}}^{u^3/3} + c$$

Some integrals whose integrands involve quantities raised to powers cannot be found by the General Power Rule. Consider the two integrals

$$\int x(x^2+1)^2\,dx \quad \text{and} \quad \int (x^2+1)^2\,dx.$$

The substitution $u = x^2 + 1$ works in the first integral but not in the second. In the second, the substitution fails because the integrand lacks the factor x needed for du. Fortunately, *for this particular integral*, you can expand the integrand as $(x^2+1)^2 = x^4 + 2x^2 + 1$ and use the (simple) Power Rule to integrate each term.

Integration by Substitution

EXAMPLE

$$\int (2x+3)^7\,dx = \frac{1}{2}\int 2(2x+3)^7\,dx$$

$$= \frac{1}{2}\,\frac{(2x+3)^8}{8} + c$$

$$= \frac{(2x+3)^8}{16} + c$$

Integration by Substitution

EXAMPLE

$$\int x(x^2+1)^4 dx$$

$= \frac{1}{2}\int 2x(x^2+1)^4 dx$

$= \frac{1}{2}\frac{(x^2+1)^5}{5} + c$

$= \frac{(x^2+1)^5}{10} + c$

Integration by Substitution

EXAMPLE

$$\int 3x^2\sqrt{x^3+5}\,dx$$

$= \int 3x^2(x^3+5)^{1/2}\, dx$

$= \frac{(x^3+5)^{3/2}}{3/2}\, dx$

$= \frac{2(x^3+5)^{3/2}}{3} + c$

Integration by Substitution

f นอก $= \frac{1}{f \text{ ใน}}$

EXAMPLE

$\boxed{\frac{d}{dx}[\ln u] = \frac{1}{u}\frac{du}{dx} = \frac{u'}{u}, \quad u > 0}$ $\boxed{\int \frac{1}{u}du = \ln|u| + C}$

$$\int \frac{x}{3x^2+4}dx$$

$= \frac{1}{6}\int \frac{6x}{3x^2+4}\, dx$

$= \frac{1}{6}\ln|3x^2+4| + c$

with a change variable : $u = 3x^2 + 4$ AND $du = 6x\,dx$

$= \frac{1}{6}\int \frac{1}{3x^2+4} 6x\, dx$

$= \frac{1}{6}\int \frac{1}{u}\, du$

$= \frac{1}{6}\ln|u| + c$

$= \frac{1}{6}\ln|3x^2+4| + c$

sin = -cos

Integration by Substitution

EXAMPLE

$$\int \sin(3x+2)\, dx$$

$= \frac{1}{3}\int 3\sin(3x+2)\, dx$

$= \frac{-\cos(3x+2)}{3} + c$

Integration by Substitution

EXAMPLE $\int \sin x \cos x \, dx$?

Let $u = \sin x \rightarrow du = \cos x \, dx$

$\int u \, du = \frac{u^2}{2} + c$

$= \frac{(\sin x)^2}{2} + c$

#Method 1

#Method 2

Let $u = \cos x \rightarrow du = -\sin x \, dx$

$\int -u \, du = -\frac{u^2}{2} + c$

$= -\frac{(\cos x)^2}{2} + c$

#Method 3

$\sin(2x) = 2(\sin x)(\cos x)$

Since $\sin x \cos x = (\frac{1}{2}) \sin(2x)$

$\int (\sin x)(\cos x) \, dx = \int (\frac{1}{2}) \sin(2x) \, dx$

$= -(\frac{1}{4}) \cos(2x) + c$

Change of Variables for Definite Integrals

When using u-substitution with a definite integral, it is often convenient to determine the limits of integration for the variable u rather than to convert the antiderivative back to the variable x and evaluate at the original limits. This change of variables is stated explicitly in the next theorem.

THEOREM 4.15 CHANGE OF VARIABLES FOR DEFINITE INTEGRALS

If the function $u = g(x)$ has a continuous derivative on the closed interval $[a, b]$ and f is continuous on the range of g, then

$$\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

EXAMPLE 8 Change of Variables

Evaluate $\int_0^1 x(x^2 + 1)^3 \, dx$.

Solution To evaluate this integral, let $u = x^2 + 1$. Then, you obtain

$u = x^2 + 1 \Rightarrow du = 2x \, dx.$

Before substituting, determine the new upper and lower limits of integration.

Lower Limit	*Upper Limit*
When $x = 0$, $u = 0^2 + 1 = 1$.	When $x = 1$, $u = 1^2 + 1 = 2$.

Now, you can substitute to obtain

$$\int_0^1 x(x^2 + 1)^3 \, dx = \frac{1}{2} \int_0^1 (x^2 + 1)^3 (2x) \, dx$$

Integration limits for x

$$= \frac{1}{2} \int_1^2 u^3 \, du$$

Integration limits for u

$$= \frac{1}{2} \left[\frac{u^4}{4} \right]_1^2$$

$$= \frac{1}{2} \left(4 - \frac{1}{4} \right)$$

$$= \frac{15}{8}.$$

Try rewriting the antiderivative $\frac{1}{2}(u^4/4)$ in terms of the variable x and evaluate the definite integral at the original limits of integration, as shown.

$$\frac{1}{2} \left[\frac{u^4}{4} \right]_1^2 = \frac{1}{2} \left[\frac{(x^2 + 1)^4}{4} \right]_0^1$$

$$= \frac{1}{2} \left(4 - \frac{1}{4} \right) = \frac{15}{8}$$

Notice that you obtain the same result. ■

EXAMPLE 9 Change of Variables

find x in terms of u

Evaluate $A = \int_1^5 \frac{x}{\sqrt{2x - 1}} \, dx.$?

let $u = \sqrt{2x - 1}$

$u^2 = 2x - 1$

$u^2 + 1 = 2x$

$\frac{u^2 + 1}{2} = x$

$u \, du = dx$

Lower Limit

When $x = 1$, $u = \sqrt{2 - 1} = 1$

Upper Limit

When $x = 5$, $u = \sqrt{10 - 1} = 3$

$\int_1^5 \frac{x}{\sqrt{2x - 1}} \, dx = \int_1^3 \frac{1}{u} \left(\frac{u^2 + 1}{2} \right) u \, du$

$= \frac{1}{2} \int_1^3 (u^2 + 1) \, du$

$= \frac{1}{2} \left[\frac{u^3}{3} + u \right]_1^3$

$= \frac{1}{2} (9 + 3 - \frac{1}{3} - 1)$

$= \frac{16}{3}$

Homework

Find

1) $\int_{-1}^{1}\left(6x^5 - \frac{2}{\sqrt[5]{x^3}} + 3\right)dx$

2) $\int_{1}^{4}|x-3|\,dx$

3) $\int \frac{(2x^2+1)}{(2x^3+3x)^2}dx$

- Write down a step-by-step solution in a piece of A4 paper (or on an electronic paper).
- **DO NOT TYPE.**
- Submit your solution in MS Team.
- Do not forget to write your name and student ID.

1) $\int_{-1}^{1}\left(6x^5 - \frac{2}{\sqrt[5]{x^3}} + 3\right)dx$

$= \int_{-1}^{1}\left(\frac{6x^6}{6} - \frac{2}{x^{3/5}} + 3x\right)$

$= \left[x^6 - 2x^{-3/5} + 3x\right]_{-1}^{1}$

$= \left[x^6 - \frac{2x^{2/5}}{2/5} + 3x\right]_{-1}^{1}$

$= \left[x^6 - \frac{10}{2}x^{2/5} + 3x\right]_{-1}^{1}$

$= \left[x^6 - 5x^{2/5} + 3x\right]_{-1}^{1}$

① แทนค่า 1 $= 1^6 - 5(1)^{2/5} + 3(1)$

$= 1 - 5\sqrt[5]{1^2} + 3$

② แทนค่า -1 $= (-1)^6 - 5\sqrt[5]{-1^2} + 3(-1)$

$= 1 - 5\sqrt[5]{-1^2} - 3$

① - ② $= \left[1 - \cancel{5\sqrt[5]{1^2}} + 3\right] - \left[1 - \cancel{5\sqrt[5]{1^2}} - 3\right]$

$= 3 - (-3)$

$= 6$ #

2) $\int_{1}^{4}|x-3|\,dx$

if $3 \geq x < 4$: $\int_{1}^{4}(x-3)\,dx = \left[\frac{x^2}{2} - 3x\right]_{1}^{4}$

① แทนค่า 4 $= \left[\frac{4^2}{2} - 3(4)\right]$

$= \frac{\cancel{16}^{8}}{\cancel{2}_{1}} - 12 = -4$

② แทนค่า 1 $= \left[\frac{1^2}{2} - 3(1)\right]$

$= \frac{1}{2} - 3 = -\frac{5}{2}$

① - ② $= -4 - \left(-\frac{5}{2}\right) = -4 + \frac{5}{2} = -\frac{3}{2}$ #

if $3 > x \geq 1$: $\int_{1}^{4}(-x+3)\,dx = \left[\frac{-x^2}{2} + 3x\right]$

① แทนค่า 4 $= \frac{-4^2}{2} + 3(4)$

$= -\frac{\cancel{16}^{8}}{\cancel{2}_{1}} + 12$

$= -8 + 12 = 4$

② แทนค่า 1 $= \frac{-1^2}{2} + 3(1)$

$= -\frac{1}{2} + 3 = \frac{5}{2}$

① - ② $= 4 - \frac{5}{2} = \frac{3}{2}$ #

Homework

Find

1) $\int_{-1}^{1}\left(6x^5 - \frac{2}{\sqrt[5]{x^3}} + 3\right)dx$

2) $\int_{1}^{4}|x-3|\,dx$

3) $\int \frac{(2x^2+1)}{(2x^3+3x)^2}\,dx$

- Write down a step-by-step solution in a piece of A4 paper (or on an electronic paper).
- **DO NOT TYPE.**
- Submit your solution in MS Team.
- Do not forget to write your name and student ID.

3) $\int \frac{(2x^2+1)}{(2x^3+3x)^2}\,dx$

u^2

$= \frac{1}{3}\int \frac{3(2x^2+1)}{(2x^3+3x)^2}\,dx$

$= \frac{1}{3}\int (2x^3+3x)^{-2}\,\underbrace{3(2x^2+1)\,dx}_{du}$

$= \frac{1}{3}\left[\frac{(2x^3+3x)^{-1}}{-1}\right] = \frac{1}{3}\left[\frac{-1}{(2x^3+3x)}\right]$

$= \frac{-1}{3(2x^3+3x)} = \frac{-1}{6x^3+9x}$ #